

Week 5 STAT 270: Class Notetake

- Increase in investment or focus on STEM vs humanities
 - Not usually including multiple perspectives in data or studies that is displayed in humanities
 - Prioritization of “critical thinking”
 - Handicap to society to not have a lot of the understandings that come from humanities / leads to use of artificial intelligence to supply these skills in necessary (interpersonal) situations
- Visualizations not inherently neutral because data is collected based on bias of inspiration of data
 - How to define neutrality in data collection when data is collected and analyzed to prove a certain point
 - Emotion is easily influenced and manipulation of it can be applied to marketing / unethical
- Can't truly make visualizations neutral because of the presentation choices that will always be a component of conveying data (color, font size, emphasis within text)
- Parascopic plot vs.
- Subject matter affects neutrality because of inherent difference in opinion
 - ex/ gun death evoking emotions
 - Lots of biases within every subject matter
 - Can this be any issue someone cares about?
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- All issues have a human component
 - Non human focused issues ex/ plants or geology
 - Still have underlying relation to human existence (plants/ environment)
- *Must think about the context in which data collection happens
- Data visualization
 - Missing context
 - Cultural and social context cannot be easily captured
- Hard to be neutral because of intention of data collection
 - Visualization model created with purpose for communication of issue of interest
 - Artistic choices
 - Colors can shape and help communicate something
 - Components can be seen as neutral but underlyingly are not
- Theme Classic
 - Zero gridlines
 - Clean slate
 - However can be harder to read (data to ink ratio)
 - Reduces artistic imprint
- Who is harmed and who benefits
 - Partner ships with those affected by study / results of study
- Plaque to learn about history in museum Artist Plaque background / Pair statisticians background with data to convey lived experiences to provide transparency of background and intention
- Explaining why stylistic choices and components are utilized
 - Convey Story behind choices
 - Why is this the storytelling

- Ex/ why area plot is used
- Rather than looking at one side of an argument considering all perspectives
- In scientific statistics
 - Must be repeatable and reproducible
 - Must have open data
 - Could require that anyone writing scientific paper must write certain statisticians backgrounds and interest or intention behind data collection
 - Really what do we want to know or distinguish what information is important to increase transparency (deeper intellectual reflection about who you are and life influences)
 - Positionality (humanities based analysis) How any aspects of lived experiences affects the research you conduct
 - I come from this background, I am doing this analysis to assess ____.
 - Can this continue to contribute to non-neutrality because some may interpret results , could harm what paper is trying to claim
 - Could increase susceptibility of polarization
 - Will claim what it claims whether there is author transparency or not
 - Those will always questions findings dependent on personal world views
 - May open the door to possibility of rejection of reading paper earlier than they would have ordinarily
- Responsibility of reader
 - Utilising data for the purpose of confirming or disproving belief already established (confirmation bias)
- This objectivity can be uncomfortable, but there is no accountability to own how perspective comes into research and acknowledge humanness (encompasses emotion inherently)
 - Hard to acknowledge influence or possible biases
 - May be uncomfortable to have people read about you in a more personal way
- Stem typically taught without multiple perspectives
 - Statistical model chosen influenced by data in attempt to make it as neutral as possible
 - Ex/ political science encompasses multiple perspectives to eliminate biases in single perspective given
- Opportunity to make space for how data is collected or interpreted based on different cultures
 - Ex/ introductory statistics textbook written solely by indigenous authors
- Plot labeling can cause viscosity for both author and consumer and influence interpretation dually
- When you don't say something it does not mean that it is actual neutrality
 - What is said when nothing is said?
- When looking at individual sources data can be connected to humanity
 - Humanity helps tell a unique story that can be emotional or deeply personal

Increasing humanization of statistics reporting!

